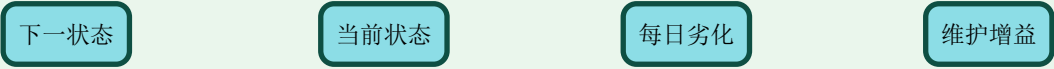


Step 1: 状态转移方程 （纳入问题一四项指标 k_0, β_s, G, α ）

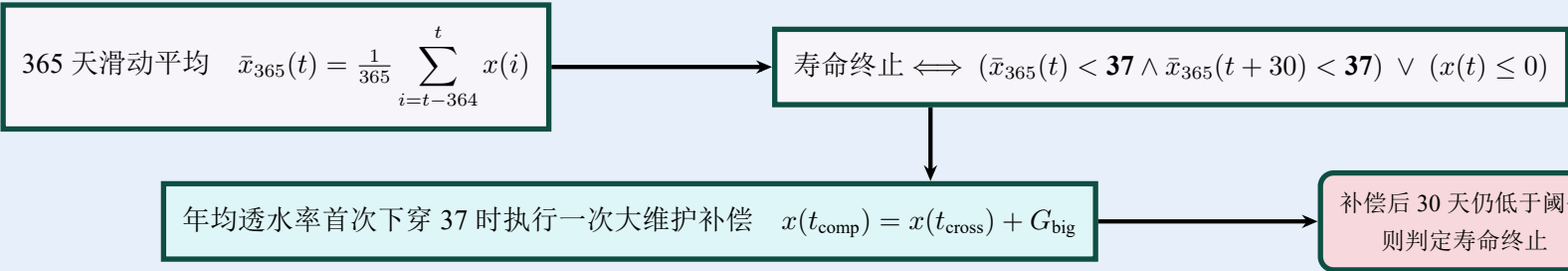


$$x(t+1) = x(t) - D(t) + G(t)$$

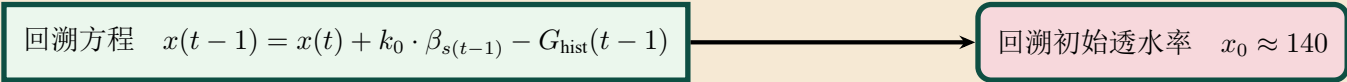
$$D(t) = k_0 \cdot \beta_{s(t)} \cdot (1 + \alpha_{\text{mid}})^{N_{\text{mid}}(t)} \cdot (1 + \alpha_{\text{big}})^{N_{\text{big}}(t)}$$

$$G(t) = \begin{cases} G_{\text{mid}}, & \text{第}t\text{天中维护} \\ G_{\text{big}}, & \text{第}t\text{天大维护} \\ 0, & \text{无维护} \end{cases}$$

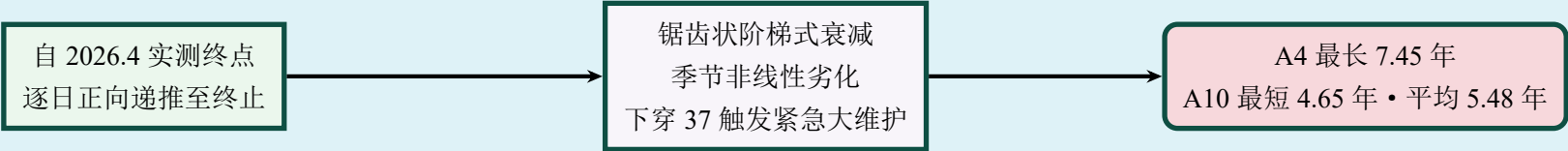
Step 2: 寿命终止判定



Step 3: 逆向回溯初始状态 （2022—2024）



Step 4: 正向寿命预测



Step 5: 可靠性验证

